

Institutional Complexity Analysis

Applying Economic Complexity Methods to V-Dem Governance Data

Institutional Complexity Project

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Introduction

Economic complexity theory, developed by Hidalgo and Hausmann (2009), provides a powerful framework for understanding why some countries are wealthier than others. The core insight is elegant: countries that can produce many products – especially products that few other countries can produce – tend to be more economically sophisticated.

Here we apply this same logic to **institutions and governance**. Using data from the Varieties of Democracy (V-Dem) project, we ask: which countries have built complex institutional portfolios? Which institutional features are “rare” – present only in countries with deep institutional capacity? And how do institutions cluster together in an “institution space”?

The analogy works as follows:

Economic Complexity	Institutional Complexity
Countries export products	Countries have institutional features
Revealed Comparative Advantage	Binarized V-Dem indicators
Product Complexity Index (PCI)	Institutional Feature Complexity (IFC)
Economic Complexity Index (ECI)	Institutional Complexity Index (ICI)
Product Space	Institution Space

Setup and Data

Load packages

```
library(tidyverse)
library(vdemdata)
library(uwot)
library(ggrepel)
library(countrycode)
library(WDI)
library(pheatmap)
library(igraph)
library(RColorBrewer)
library(knitr)
library(DT)
```

Load V-Dem data

We use the `vdemdata` package which provides the full V-Dem dataset. We'll work with the most recent year that has good coverage.

```
vdem_raw <- vdem

# Check the most recent years and coverage
year_coverage <- vdem_raw %>%
  filter(year >= 2020) %>%
  group_by(year) %>%
  summarise(
    n_countries = n_distinct(country_name),
    .groups = "drop"
  )

kable(year_coverage, caption = "Country coverage by year (2020+)")
```

Table 2: Country coverage by year (2020+)

year	n_countries
2020	179
2021	179
2022	179
2023	179
2024	179
2025	179

```
# Use the most recent year with substantial coverage
latest_year <- year_coverage %>%
  filter(n_countries >= 150) %>%
  pull(year) %>%
  max()

cat("Selected year:", latest_year, "\n")
```

Selected year: 2025

Select institutional indicators

We curate a set of institutional and governance indicators spanning multiple domains. V-Dem indicators come in several forms:

- **Ordinal (0-4):** Expert-coded assessments (e.g., `v2x_*` indices are often interval, but many component indicators like `v2cl*`, `v2ju*` are ordinal)
- **Interval (0-1):** Measurement-model aggregated indices
- **Continuous:** Some structural variables

We focus on indicators that capture the *presence* or *quality* of specific institutional features.

```
# Define our indicator set with categories
indicator_info <- tribble(
  ~code, ~category, ~description,
  # Rule of Law
  "v2x_rule", "Rule of Law", "Rule of law index",
  "v2xcl_rol", "Rule of Law", "Equality before the law and individual liberty index",
  "v2clrspct", "Rule of Law", "Rigorous and impartial public administration",
  "v2cltrnslw", "Rule of Law", "Transparent laws with predictable enforcement",
  "v2clacjust", "Rule of Law", "Access to justice for men",
  "v2clacjstw", "Rule of Law", "Access to justice for women",
  "v2clsocgrp", "Rule of Law", "Social class equality in respect for civil liberties",
  "v2clrgunev", "Rule of Law", "Even geographic distribution of civil liberties",

  # Judicial Independence
  "v2x_jucon", "Judicial Independence", "Judicial constraints on the executive index",
  "v2juhcind", "Judicial Independence", "High court independence",
  "v2juncind", "Judicial Independence", "Lower court independence",
  "v2juhccomp", "Judicial Independence", "Compliance with high court decisions",
  "v2jucomp", "Judicial Independence", "Compliance with judiciary",
  "v2jureview", "Judicial Independence", "Judicial review",
  "v2juacct", "Judicial Independence", "Judicial accountability",
  "v2jupack", "Judicial Independence", "Court packing",
  "v2jupurge", "Judicial Independence", "Purges of the judiciary",
```

Bureaucratic Quality / State Capacity

"v2stcritrecadm", "State Capacity", "Criteria for appointment decisions in state administration",
"v2strenadm", "State Capacity", "Remuneration in state administration",
"v2x_pubcorr", "State Capacity", "Public sector corruption index",
"v2clrspct", "State Capacity", "Rigorous and impartial public administration",
"v2svstterr", "State Capacity", "State authority over territory",
"v2stfisci", "State Capacity", "Fiscal capacity",
"v2ststatag", "State Capacity", "State statistical agency capacity",

Property Rights

"v2xcl_prpty", "Property Rights", "Property rights index",
"v2clprptym", "Property Rights", "Property rights for men",
"v2clprptyw", "Property Rights", "Property rights for women",

Media Freedom

"v2x_freexp_altinf", "Media Freedom", "Freedom of expression and alternative sources of info index",
"v2mecenefm", "Media Freedom", "Government censorship effort -- media",
"v2meharjrn", "Media Freedom", "Harassment of journalists",
"v2meslfcen", "Media Freedom", "Media self-censorship",
"v2mebias", "Media Freedom", "Media bias",
"v2mecrit", "Media Freedom", "Print/broadcast media critical",
"v2merange", "Media Freedom", "Print/broadcast media perspectives",
"v2x_freexp", "Media Freedom", "Freedom of expression index",
"v2meaccess", "Media Freedom", "Media access",

Electoral Integrity

"v2x_polyarchy", "Electoral Integrity", "Electoral democracy index",
"v2xel_frefair", "Electoral Integrity", "Clean elections index",
"v2elfrfair", "Electoral Integrity", "Free and fair elections",
"v2elirreg", "Electoral Integrity", "Election irregularities",
"v2elintim", "Electoral Integrity", "Election government intimidation",
"v2elpeace", "Electoral Integrity", "Election other voting irregularities",
"v2elvotbuy", "Electoral Integrity", "Election vote buying",
"v2elembaut", "Electoral Integrity", "EMB autonomy",
"v2elembcap", "Electoral Integrity", "EMB capacity",
"v2elrgstry", "Electoral Integrity", "Election voter registry",
"v2x_elecoff", "Electoral Integrity", "Elected officials index",

Civil Liberties

"v2x_civlib", "Civil Liberties", "Civil liberties index",
"v2xcl_disc", "Civil Liberties", "Freedom of discussion index",
"v2clacfree", "Civil Liberties", "Freedom of academic and cultural expression",
"v2cldiscm", "Civil Liberties", "Freedom of discussion for men",
"v2cldiscw", "Civil Liberties", "Freedom of discussion for women",
"v2clfmov", "Civil Liberties", "Freedom of foreign movement",
"v2cldmovm", "Civil Liberties", "Freedom of domestic movement for men",
"v2cldmovw", "Civil Liberties", "Freedom of domestic movement for women",
"v2clkill", "Civil Liberties", "Freedom from political killings",
"v2clslavem", "Civil Liberties", "Freedom from forced labor for men",
"v2clslavef", "Civil Liberties", "Freedom from forced labor for women",
"v2cltort", "Civil Liberties", "Freedom from torture",
"v2clstown", "Civil Liberties", "State ownership of economy",
"v2clrelig", "Civil Liberties", "Freedom of religion",

Legislative Constraints

"v2x_legcon", "Legislative Constraints", "Legislative constraints on executive index",
"v2lgqstexp", "Legislative Constraints", "Legislature questions officials in practice",
"v2lginvstp", "Legislative Constraints", "Legislature investigates in practice",
"v2lgotovst", "Legislative Constraints", "Legislature opposition parties oversight",
"v2lgcrrpt", "Legislative Constraints", "Legislature corrupt activities",
"v2lgoppart", "Legislative Constraints", "Legislature opposition parties",
"v2lgfunds", "Legislative Constraints", "Legislature controls resources",
"v2lgdsadlo", "Legislative Constraints", "Legislature disbands/self-limits",
"v2lgello", "Legislative Constraints", "Legislature elections are free/fair",

Anti-Corruption

"v2x_corr", "Anti-Corruption", "Political corruption index",
"v2excrtps", "Anti-Corruption", "Executive corruption (public sector)",
"v2exthtfts", "Anti-Corruption", "Executive embezzlement",
"v2exbribe", "Anti-Corruption", "Executive bribery",
"v2lgcrrpt", "Anti-Corruption", "Legislature corrupt activities",
"v2jucorrdc", "Anti-Corruption", "Judicial corruption decision",

Association and Civil Society

"v2x_cspart", "Civil Society", "Civil society participation index",
"v2csprtcpt", "Civil Society", "Civil society participatory environment",
"v2cseeorgs", "Civil Society", "CSO entry and exit",
"v2csreprss", "Civil Society", "CSO repression",
"v2csantimv", "Civil Society", "CSO anti-system movements",
"v2x_frassoc_thick", "Civil Society", "Freedom of association (thick) index",

Academic Freedom

"v2xca_academ", "Academic Freedom", "Academic and cultural freedom index",
"v2caassemb", "Academic Freedom", "Freedom of academic exchange",
"v2cafexch", "Academic Freedom", "Freedom of academic exchange and dissemination",
"v2casurv", "Academic Freedom", "Surveillance of academic activities",

Deliberation

"v2x_delibdem", "Deliberation", "Deliberative democracy index",
"v2dlreason", "Deliberation", "Reasoned justification",
"v2dlcommon", "Deliberation", "Common good justification",
"v2dlcountr", "Deliberation", "Respect for counterarguments",
"v2dlconslt", "Deliberation", "Range of consultation",
"v2dlengage", "Deliberation", "Engaged society",

Equality

"v2x_gender", "Equality", "Women's political empowerment index",
"v2x_genpp", "Equality", "Women's civil society participation index",
"v2x_genc1", "Equality", "Women's civil liberties index",
"v2pepwrngen", "Equality", "Power distributed by gender",
"v2pepwrres", "Equality", "Power distributed by socioeconomic position",
"v2pepwrsoc", "Equality", "Power distributed by social group",

Accountability

"v2x_accountability", "Accountability", "Accountability index",
"v2x_veracc", "Accountability", "Vertical accountability index",
"v2x_horacc", "Accountability", "Horizontal accountability index",

```

    "v2x_diagacc", "Accountability", "Diagonal accountability index"
  )

# Remove duplicates (some indicators appear in multiple categories -- keep first)
indicator_info <- indicator_info %>% distinct(code, .keep_all = TRUE)

cat("Total unique indicators selected:", nrow(indicator_info), "\n")

## Total unique indicators selected: 100

cat("Categories:\n")

## Categories:
indicator_info %>% count(category, sort = TRUE) %>% print(n = 20)

## # A tibble: 14 x 2
##   category      n
##   <chr>      <int>
## 1 Civil Liberties      14
## 2 Electoral Integrity   11
## 3 Judicial Independence    9
## 4 Legislative Constraints    9
## 5 Media Freedom           9
## 6 Rule of Law             8
## 7 Civil Society           6
## 8 Deliberation            6
## 9 Equality                6
## 10 State Capacity         6
## 11 Anti-Corruption        5
## 12 Academic Freedom       4
## 13 Accountability         4
## 14 Property Rights        3

# Check which of our selected indicators actually exist in the dataset
available_codes <- intersect(indicator_info$code, names(vdem_raw))
missing_codes <- setdiff(indicator_info$code, names(vdem_raw))

if (length(missing_codes) > 0) {
  cat("Indicators not found in dataset (will be dropped):\n")
  cat(paste(missing_codes, collapse = ", "), "\n\n")
}

## Indicators not found in dataset (will be dropped):
## v2stfisci, v2ststatag, v2meaccess, v2x_legcon

indicator_info <- indicator_info %>% filter(code %in% available_codes)
cat("Indicators available:", nrow(indicator_info), "\n")

## Indicators available: 96

# Filter to selected year and extract relevant columns
vdem_filtered <- vdem_raw %>%
  filter(year == latest_year) %>%
  select(
    country_name, country_text_id, COWcode,
    e_regionpol, e_regionpol_6C,

```

```

  all_of(indicator_info$code)
) %>%
# Drop countries with too much missing data (> 50% missing)
mutate(n_missing = rowSums(is.na(across(all_of(indicator_info$code))))) %>%
filter(n_missing < ncol(across(all_of(indicator_info$code))) * 0.5) %>%
select(-n_missing)

cat("Countries retained:", nrow(vdem_filtered), "\n")

```

```
## Countries retained: 179
```

Indicator catalog

Here is the full set of institutional indicators we're working with:

```

kable(
  indicator_info %>% select(Code = code, Category = category, Description = description),
  caption = "Institutional indicators used in this analysis"
)

```

Table 3: Institutional indicators used in this analysis

Code	Category	Description
v2x_rule	Rule of Law	Rule of law index
v2xcl_rol	Rule of Law	Equality before the law and individual liberty index
v2clrspct	Rule of Law	Rigorous and impartial public administration
v2cltrnslw	Rule of Law	Transparent laws with predictable enforcement
v2clacjust	Rule of Law	Access to justice for men
v2clacjstw	Rule of Law	Access to justice for women
v2clsocgrp	Rule of Law	Social class equality in respect for civil liberties
v2clrgunev	Rule of Law	Even geographic distribution of civil liberties
v2x_jucon	Judicial Independence	Judicial constraints on the executive index
v2juhcind	Judicial Independence	High court independence
v2juncind	Judicial Independence	Lower court independence
v2juhccomp	Judicial Independence	Compliance with high court decisions
v2jucomp	Judicial Independence	Compliance with judiciary
v2jureview	Judicial Independence	Judicial review
v2juacct	Judicial Independence	Judicial accountability
v2jupack	Judicial Independence	Court packing
v2jupurge	Judicial Independence	Purges of the judiciary
v2stcritrecadm	State Capacity	Criteria for appointment decisions in state administration
v2strenadm	State Capacity	Remuneration in state administration
v2x_pubcorr	State Capacity	Public sector corruption index
v2svstterr	State Capacity	State authority over territory
v2xcl_prpty	Property Rights	Property rights index
v2clprptym	Property Rights	Property rights for men
v2clprptyw	Property Rights	Property rights for women
v2x_freexp_altinf	Media Freedom	Freedom of expression and alternative sources of info index
v2mecenefm	Media Freedom	Government censorship effort – media
v2meharjrn	Media Freedom	Harassment of journalists
v2meslfcen	Media Freedom	Media self-censorship
v2mebias	Media Freedom	Media bias
v2mecrit	Media Freedom	Print/broadcast media critical
v2merange	Media Freedom	Print/broadcast media perspectives

Code	Category	Description
v2x_freexp	Media Freedom	Freedom of expression index
v2x_polyarchy	Electoral Integrity	Electoral democracy index
v2xel_frefair	Electoral Integrity	Clean elections index
v2elfrfair	Electoral Integrity	Free and fair elections
v2elirreg	Electoral Integrity	Election irregularities
v2elintim	Electoral Integrity	Election government intimidation
v2elpeace	Electoral Integrity	Election other voting irregularities
v2elvotbuy	Electoral Integrity	Election vote buying
v2elembaut	Electoral Integrity	EMB autonomy
v2elembcap	Electoral Integrity	EMB capacity
v2elrgstry	Electoral Integrity	Election voter registry
v2x_elecoff	Electoral Integrity	Elected officials index
v2x_civlib	Civil Liberties	Civil liberties index
v2xcl_disc	Civil Liberties	Freedom of discussion index
v2clacfree	Civil Liberties	Freedom of academic and cultural expression
v2cldiscm	Civil Liberties	Freedom of discussion for men
v2cldiscw	Civil Liberties	Freedom of discussion for women
v2clfmov	Civil Liberties	Freedom of foreign movement
v2cldmovem	Civil Liberties	Freedom of domestic movement for men
v2cldmovew	Civil Liberties	Freedom of domestic movement for women
v2clkill	Civil Liberties	Freedom from political killings
v2clslavem	Civil Liberties	Freedom from forced labor for men
v2clslavef	Civil Liberties	Freedom from forced labor for women
v2cltort	Civil Liberties	Freedom from torture
v2clstown	Civil Liberties	State ownership of economy
v2clrelig	Civil Liberties	Freedom of religion
v2lgqstexp	Legislative Constraints	Legislature questions officials in practice
v2lginvstp	Legislative Constraints	Legislature investigates in practice
v2lgotovst	Legislative Constraints	Legislature opposition parties oversight
v2lgcrpt	Legislative Constraints	Legislature corrupt activities
v2lgoppart	Legislative Constraints	Legislature opposition parties
v2lgfunds	Legislative Constraints	Legislature controls resources
v2lgdsadlo	Legislative Constraints	Legislature disbands/self-limits
v2lgello	Legislative Constraints	Legislature elections are free/fair
v2x_corr	Anti-Corruption	Political corruption index
v2excrptps	Anti-Corruption	Executive corruption (public sector)
v2exthftps	Anti-Corruption	Executive embezzlement
v2exbribe	Anti-Corruption	Executive bribery
v2jucorrde	Anti-Corruption	Judicial corruption decision
v2x_cspart	Civil Society	Civil society participation index
v2csptrcpt	Civil Society	Civil society participatory environment
v2cseeorgs	Civil Society	CSO entry and exit
v2csreprss	Civil Society	CSO repression
v2csantimv	Civil Society	CSO anti-system movements
v2x_frassoc_thick	Civil Society	Freedom of association (thick) index
v2xca_academ	Academic Freedom	Academic and cultural freedom index
v2caasemb	Academic Freedom	Freedom of academic exchange
v2cafexch	Academic Freedom	Freedom of academic exchange and dissemination
v2casurv	Academic Freedom	Surveillance of academic activities
v2x_delibdem	Deliberation	Deliberative democracy index
v2dlreason	Deliberation	Reasoned justification
v2dlcommon	Deliberation	Common good justification

Code	Category	Description
v2dlcountr	Deliberation	Respect for counterarguments
v2dlconslt	Deliberation	Range of consultation
v2dlengage	Deliberation	Engaged society
v2x_gender	Equality	Women's political empowerment index
v2x_genpp	Equality	Women's civil society participation index
v2x_gencl	Equality	Women's civil liberties index
v2pepwrngen	Equality	Power distributed by gender
v2pepwrres	Equality	Power distributed by socioeconomic position
v2pepwrsoc	Equality	Power distributed by social group
v2x_accountability	Accountability	Accountability index
v2x_veracc	Accountability	Vertical accountability index
v2x_horacc	Accountability	Horizontal accountability index
v2x_diagacc	Accountability	Diagonal accountability index

Binarization

The key step in economic complexity is constructing a binary **country** $\times$ **feature** matrix. In trade data, this is done via Revealed Comparative Advantage (RCA). Here we adapt the approach to V-Dem indicators.

Strategy

V-Dem indicators come in two main forms:

1. **Ordinal (0-4)**: Expert-coded assessments where higher = more democratic/better governance. We binarize at ≥ 3 , meaning the feature is “present” when a country scores at least 3 out of 4.
2. **Interval (0-1)**: Aggregated indices from measurement models. We use an **RCA-like approach**: a country “has” the feature if its value exceeds the global mean for that indicator.

```
# Determine type for each indicator based on its range in the data
classify_indicator <- function(values) {
  values <- na.omit(values)
  rng <- range(values)

  # If max is close to 1 and min is close to 0, it's likely interval

  if (rng[2] <= 1.1 & rng[1] >= -0.1) {
    return("interval")
  }
  # If values look like integers in 0-4 range, ordinal
  if (rng[2] <= 5 & all(abs(values - round(values)) < 0.01)) {
    return("ordinal")
  }
  # Default: treat as interval with RCA approach
  return("interval")
}

indicator_types <- tibble(
  code = indicator_info$code,
  type = map_chr(indicator_info$code, ~classify_indicator(vdem_filtered[[.x]]))
)

cat("Indicator types:\n")
```

```

## Indicator types:
count(indicator_types, type)

## # A tibble: 1 x 2
##   type      n
##   <chr>    <int>
## 1 interval  96

# Build the binary matrix
binary_df <- vdem_filtered %>% select(country_name, all_of(indicator_info$code))

for (ind in indicator_info$code) {
  ind_type <- indicator_types$type[indicator_types$code == ind]
  values <- binary_df[[ind]]

  if (ind_type == "ordinal") {
    binary_df[[ind]] <- as.integer(values >= 3)
  } else {
    # RCA-like: above global mean
    global_mean <- mean(values, na.rm = TRUE)
    binary_df[[ind]] <- as.integer(values >= global_mean)
  }
}

# Replace remaining NAs with 0 (absence)
binary_df <- binary_df %>%
  mutate(across(all_of(indicator_info$code), ~replace_na(.x, 0L)))

# Create the Mcp matrix (countries x institutional features)
Mcp <- binary_df %>%
  column_to_rownames("country_name") %>%
  as.matrix()

cat("Mcp matrix dimensions:", nrow(Mcp), "countries x", ncol(Mcp), "institutional features\n")

## Mcp matrix dimensions: 179 countries x 96 institutional features
cat("Matrix density (fraction of 1s):", round(mean(Mcp), 3), "\n")

## Matrix density (fraction of 1s): 0.552

The density of the matrix tells us something important. If it's too high (> 0.8) or too low (< 0.1), the complexity calculations won't be very informative. A density around 0.3-0.6 is typical for complexity analyses.

# Summary statistics
diversity <- rowSums(Mcp)
ubiquity <- colSums(Mcp)

cat("\nDiversity (# features per country):\n")

##
## Diversity (# features per country):
cat("  Mean:", round(mean(diversity), 1), "\n")

##   Mean: 53

```

```
cat("  Range:", min(diversity), "-", max(diversity), "\n")
```

```
##    Range: 2 - 94
```

```
cat("\nUbiquity (# countries per feature):\n")
```

```
##
```

```
## Ubiquity (# countries per feature):
```

```
cat("  Mean:", round(mean(ubiquity), 1), "\n")
```

```
##    Mean: 98.9
```

```
cat("  Range:", min(ubiquity), "-", max(ubiquity), "\n")
```

```
##    Range: 71 - 145
```

Complexity Calculation

Diversity and Ubiquity (0th order)

The most basic measures: **diversity** is how many institutional features a country has; **ubiquity** is how many countries share a given feature.

```
# Diversity: number of features each country has
```

```
kc0 <- rowSums(Mcp)
```

```
# Ubiquity: number of countries that have each feature
```

```
kp0 <- colSums(Mcp)
```

```
diversity_df <- tibble(  
  country = names(kc0),  
  diversity = as.numeric(kc0)  
) %>%  
  arrange(desc(diversity))
```

```
# Top and bottom by diversity
```

```
cat("Top 10 countries by diversity (# institutional features):\n")
```

```
## Top 10 countries by diversity (# institutional features):
```

```
head(diversity_df, 10) %>% kable()
```

country	diversity
France	94
Germany	94
Spain	94
Sweden	93
Australia	93
Netherlands	93
Austria	93
Czechia	93
Denmark	93
Estonia	93

```
cat("\nBottom 10 countries by diversity:\n")
```

```
##
```

```
## Bottom 10 countries by diversity:
```

```
tail(diversity_df, 10) %>% kable()
```

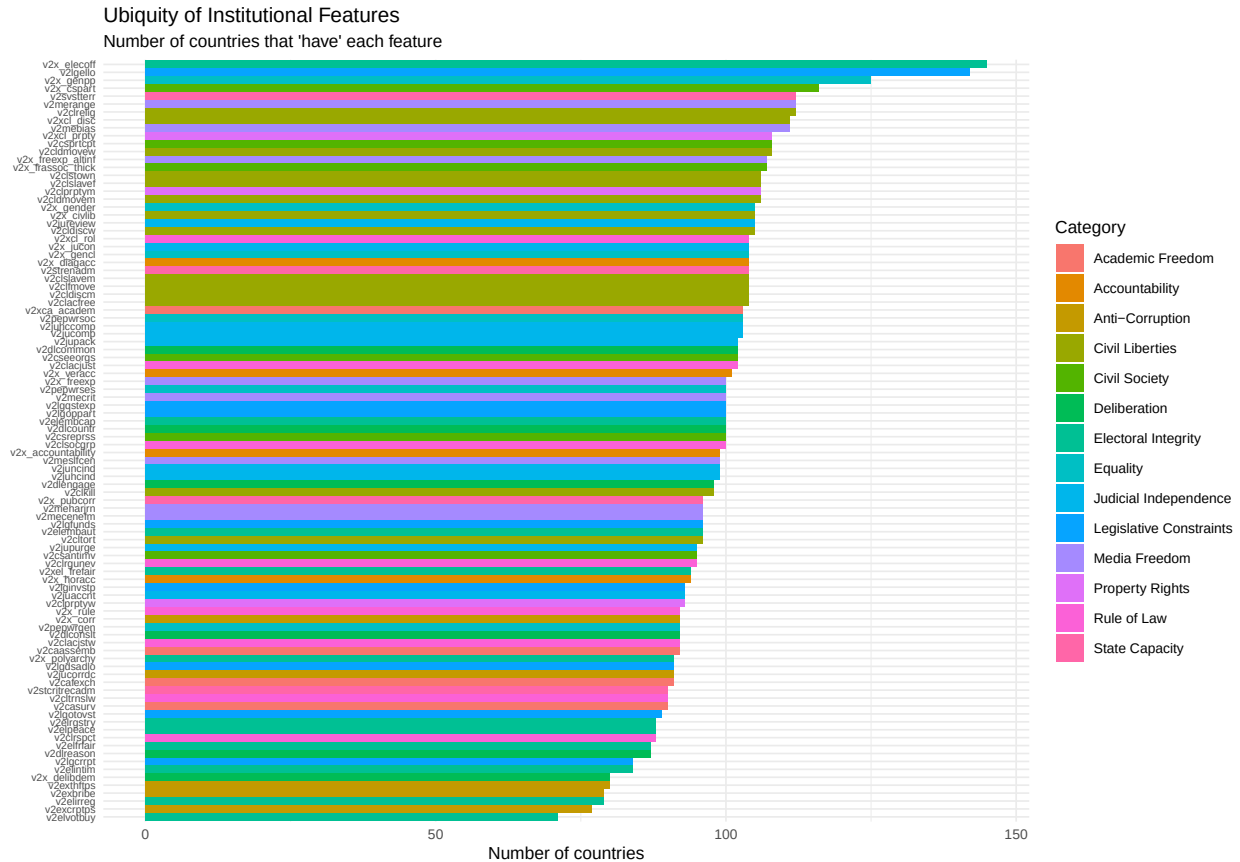
country	diversity
Syria	8
Nicaragua	7
Cambodia	6
Equatorial Guinea	6
Eritrea	5
Burma/Myanmar	4
Yemen	4
Sudan	4
South Sudan	3
Afghanistan	2

```
ubiquity_df <- tibble(  
  indicator = names(kp0),  
  ubiquity = as.numeric(kp0)
```

```
) %>%
```

```
  left_join(indicator_info, by = c("indicator" = "code")) %>%  
  arrange(ubiquity)
```

```
ggplot(ubiquity_df, aes(x = reorder(indicator, ubiquity), y = ubiquity, fill = category)) +  
  geom_col() +  
  coord_flip() +  
  labs(  
    title = "Ubiquity of Institutional Features",  
    subtitle = "Number of countries that 'have' each feature",  
    x = NULL, y = "Number of countries",  
    fill = "Category"  
  ) +  
  theme_minimal(base_size = 10) +  
  theme(axis.text.y = element_text(size = 6))
```



Method of Reflections

The Method of Reflections iteratively refines diversity and ubiquity. At each iteration, a country's complexity becomes the average complexity of the features it has, weighted by how complex those features are – and vice versa. This creates a recursive, self-consistent measure.

```
# Method of Reflections
n_iterations <- 20

# Initialize
kc <- matrix(0, nrow = nrow(Mcp), ncol = n_iterations + 1)
kp <- matrix(0, nrow = ncol(Mcp), ncol = n_iterations + 1)
rownames(kc) <- rownames(Mcp)
rownames(kp) <- colnames(Mcp)

kc[, 1] <- kc0
kp[, 1] <- kp0

for (n in 1:n_iterations) {
  # Update country complexity: average ubiquity-weighted feature complexity
  for (c in 1:nrow(Mcp)) {
    features <- which(Mcp[c, ] == 1)
    if (length(features) > 0 & kc0[c] > 0) {
      kc[c, n + 1] <- (1 / kc0[c]) * sum(kp[features, n])
    }
  }
}
```

```

# Update feature complexity: average diversity-weighted country complexity
for (p in 1:ncol(Mcp)) {
  countries <- which(Mcp[, p] == 1)
  if (length(countries) > 0 & kp0[p] > 0) {
    kp[p, n + 1] <- (1 / kp0[p]) * sum(kc[countries, n])
  }
}
}

# ECI from Method of Reflections (using even iterations for countries)
# The complexity information is in the variance of the even-iteration values
kc_even <- kc[, n_iterations + 1] # Last even iteration
eci_mor <- (kc_even - mean(kc_even)) / sd(kc_even)

kp_even <- kp[, n_iterations + 1]
pci_mor <- (kp_even - mean(kp_even)) / sd(kp_even)

cat("Method of Reflections converged after", n_iterations, "iterations.\n")

```

Method of Reflections converged after 20 iterations.

Eigenvalue Method

The eigenvalue method is more elegant and mathematically equivalent to the Method of Reflections at convergence. We construct the matrix $\tilde{M}_{cc'}$ and take its second eigenvector.

$$\tilde{M}_{cc'} = \sum_p \frac{M_{cp}M_{c'p}}{k_{c,0}k_{p,0}}$$

```

# Build Mcc_tilde
# Mcc_tilde = (1/kc0) * Mcp %%% diag(1/kp0) %%% t(Mcp)

# Handle zeros
kc0_safe <- ifelse(kc0 == 0, 1, kc0)
kp0_safe <- ifelse(kp0 == 0, 1, kp0)

Mcc_tilde <- diag(1 / kc0_safe) %%% Mcp %%% diag(1 / kp0_safe) %%% t(Mcp)
rownames(Mcc_tilde) <- rownames(Mcp)
colnames(Mcc_tilde) <- rownames(Mcp)

# Eigen decomposition
eig <- eigen(Mcc_tilde)

# The first eigenvector is trivial (all same sign), use second
eci_eigen_raw <- Re(eig$vectors[, 2])
names(eci_eigen_raw) <- rownames(Mcp)

# Standardize
eci_eigen <- (eci_eigen_raw - mean(eci_eigen_raw)) / sd(eci_eigen_raw)

# Ensure sign convention: positive ECI = more complex
# Check correlation with diversity -- complex countries should have higher diversity
if (cor(eci_eigen, kc0) < 0) {

```

```

  eci_eigen <- -eci_eigen
}

# Similarly for PCI via Mpp_tilde
Mpp_tilde <- diag(1 / kp0_safe) %*% t(Mcp) %*% diag(1 / kc0_safe) %*% Mcp
rownames(Mpp_tilde) <- colnames(Mcp)
colnames(Mpp_tilde) <- colnames(Mcp)

eig_p <- eigen(Mpp_tilde)
pci_eigen_raw <- Re(eig_p$vectors[, 2])
names(pci_eigen_raw) <- colnames(Mcp)
pci_eigen <- (pci_eigen_raw - mean(pci_eigen_raw)) / sd(pci_eigen_raw)

if (cor(pci_eigen, kp0, use = "complete.obs") > 0) {
  # More ubiquitous features should be LESS complex
  pci_eigen <- -pci_eigen
}

cat("Correlation between MoR and eigenvalue ECI:",
    round(cor(eci_mor, eci_eigen), 3), "\n")

```

```
## Correlation between MoR and eigenvalue ECI: 1
```

Results: Institutional Complexity Index

We use the eigenvalue method as our primary measure (it's more robust), with the Method of Reflections as a check.

```

eci_df <- tibble(
  country = names(eci_eigen),
  ICI = eci_eigen,
  ICI_MoR = eci_mor,
  diversity = kc0
) %>%
  arrange(desc(ICI))

cat("Top 20 countries by Institutional Complexity Index:\n")

```

```
## Top 20 countries by Institutional Complexity Index:
```

```

eci_df %>%
  head(20) %>%
  mutate(across(where(is.numeric), ~round(.x, 3))) %>%
  kable()

```

country	ICI	ICI_MoR	diversity
Oman	0.840	0.840	21
United Arab Emirates	0.717	0.716	21
The Gambia	0.707	0.707	84
Bhutan	0.703	0.703	75
Botswana	0.695	0.695	77
Cyprus	0.689	0.689	85
Kuwait	0.688	0.688	25
Italy	0.680	0.680	86

country	ICI	ICI_MoR	diversity
Kosovo	0.673	0.673	71
Sao Tome and Principe	0.664	0.664	80
Suriname	0.664	0.664	85
Colombia	0.663	0.663	71
Mauritius	0.654	0.654	86
Switzerland	0.652	0.652	91
Cape Verde	0.651	0.651	87
Portugal	0.650	0.650	85
Maldives	0.650	0.650	70
Costa Rica	0.649	0.649	91
Moldova	0.648	0.648	75
Uruguay	0.647	0.647	88

```
cat("\nBottom 20 countries by Institutional Complexity Index:\n")
```

```
##
```

```
## Bottom 20 countries by Institutional Complexity Index:
```

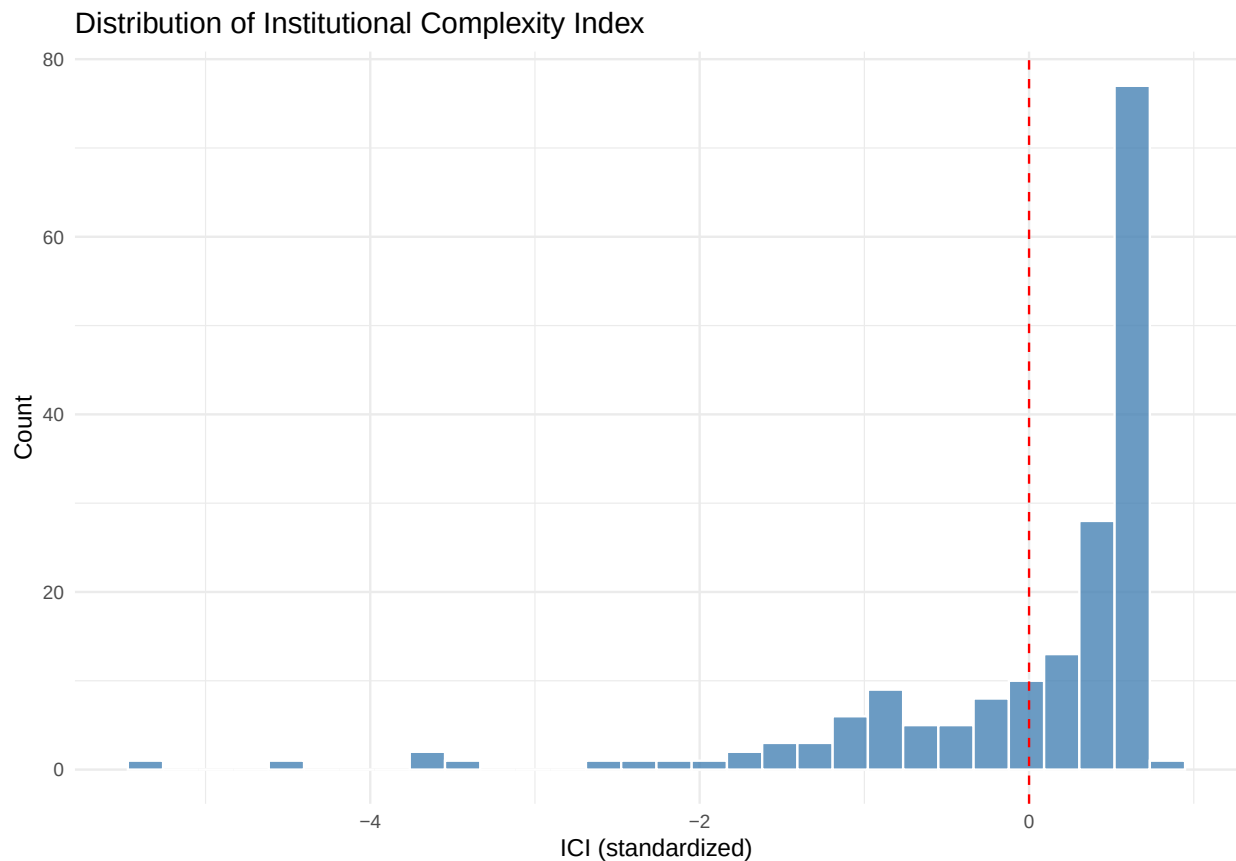
```
eci_df %>%
  tail(20) %>%
  mutate(across(where(is.numeric), ~round(.x, 3))) %>%
  kable()
```

country	ICI	ICI_MoR	diversity
Chad	-1.106	-1.106	12
Eswatini	-1.112	-1.112	11
Tajikistan	-1.171	-1.171	12
Türkiye	-1.208	-1.208	14
Russia	-1.225	-1.225	11
Burundi	-1.299	-1.299	12
Syria	-1.457	-1.457	8
Djibouti	-1.533	-1.533	12
Republic of the Congo	-1.559	-1.559	10
Turkmenistan	-1.631	-1.632	9
Uzbekistan	-1.756	-1.756	9
Nicaragua	-1.972	-1.972	7
Cambodia	-2.197	-2.197	6
Eritrea	-2.364	-2.364	5
Equatorial Guinea	-2.542	-2.542	6
Sudan	-3.378	-3.378	4
Burma/Myanmar	-3.562	-3.562	4
Yemen	-3.687	-3.687	4
South Sudan	-4.450	-4.450	3
Afghanistan	-5.365	-5.365	2

```
ggplot(eci_df, aes(x = ICI)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white", alpha = 0.8) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
  labs(
    title = "Distribution of Institutional Complexity Index",
    x = "ICI (standardized)", y = "Count"
```



```
) +  
theme_minimal(base_size = 14)
```



Results: Institutional Feature Complexity

Which institutional features are most “complex” – i.e., found predominantly in institutionally sophisticated countries?

```
pci_df <- tibble(  
  indicator = names(pci_eigen),  
  IFC = pci_eigen,  
  ubiquity = kp0  
) %>%  
  left_join(indicator_info, by = c("indicator" = "code")) %>%  
  arrange(desc(IFC))  
  
cat("Top 20 most complex institutional features:\n")
```

Top 20 most complex institutional features:

```
pci_df %>%  
  head(20) %>%  
  select(indicator, description, category, IFC, ubiquity) %>%  
  mutate(IFC = round(IFC, 3)) %>%  
  kable()
```

indicator	description	category	IFC	ubiquity
v2exthtfts	Executive embezzlement	Anti-Corruption	0.891	80
v2x_rule	Rule of law index	Rule of Law	0.871	92
v2excrptps	Executive corruption (public sector)	Anti-Corruption	0.856	77
v2x_delibdem	Deliberative democracy index	Deliberation	0.815	80
v2elintim	Election government intimidation	Electoral Integrity	0.776	84
v2clrspct	Rigorous and impartial public administration	Rule of Law	0.751	88
v2x_polyarchy	Electoral democracy index	Electoral Integrity	0.747	91
v2xel_frefair	Clean elections index	Electoral Integrity	0.716	94
v2elrgstry	Election voter registry	Electoral Integrity	0.702	88
v2elfrfair	Free and fair elections	Electoral Integrity	0.678	87
v2cltort	Freedom from torture	Civil Liberties	0.669	96
v2elirreg	Election irregularities	Electoral Integrity	0.629	79
v2casurv	Surveillance of academic activities	Academic	0.607	90
		Freedom		
v2x_accountability	Accountability index	Accountability	0.602	99
v2cltrnslw	Transparent laws with predictable enforcement	Rule of Law	0.597	90
v2x_freexp	Freedom of expression index	Media Freedom	0.597	100
v2xcl_rol	Equality before the law and individual liberty index	Rule of Law	0.589	104
v2elembaut	EMB autonomy	Electoral Integrity	0.581	96
v2jucorrdc	Judicial corruption decision	Anti-Corruption	0.577	91
v2x_horacc	Horizontal accountability index	Accountability	0.566	94

```
cat("\nBottom 20 (most ubiquitous/least complex) features:\n")
```

```
##
```

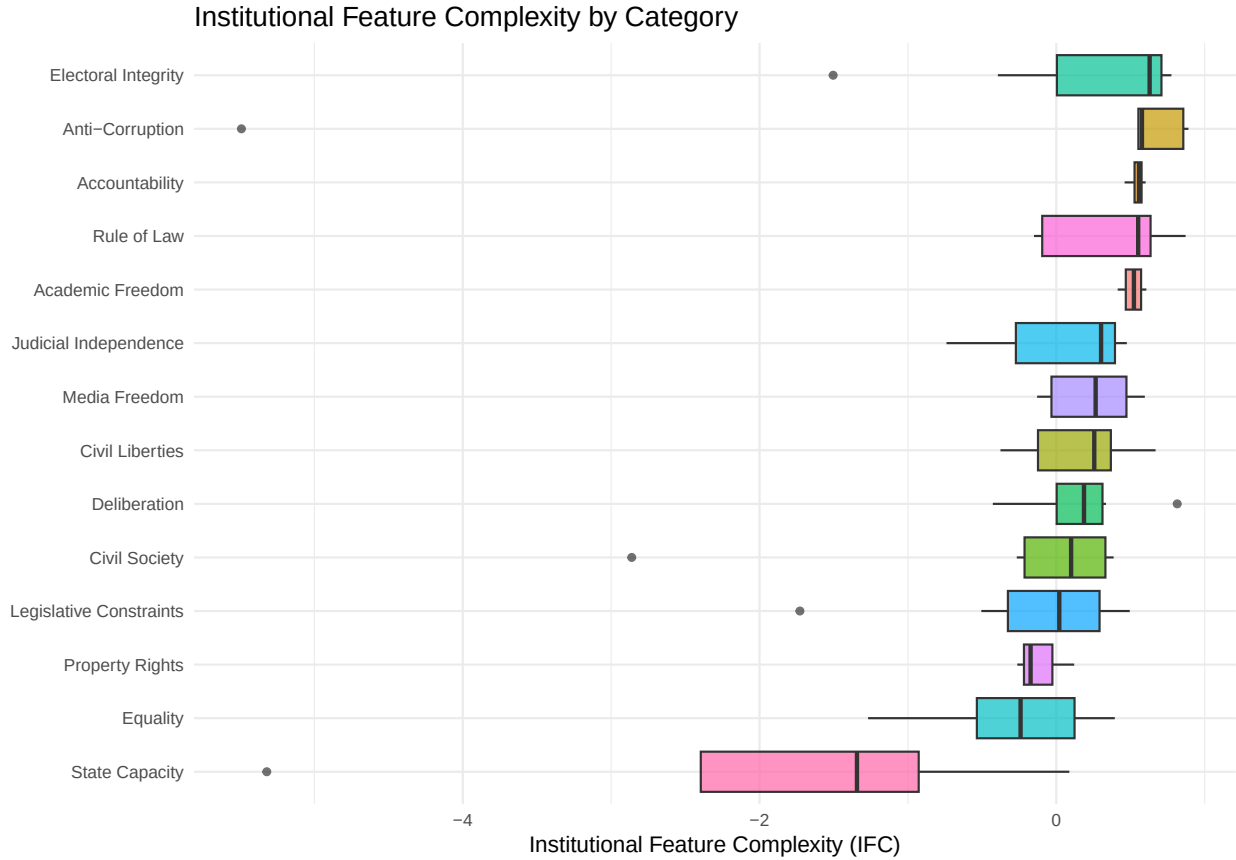
```
## Bottom 20 (most ubiquitous/least complex) features:
```

```
pci_df %>%
  tail(20) %>%
  select(indicator, description, category, IFC, ubiquity) %>%
  mutate(IFC = round(IFC, 3)) %>%
  kable()
```

indicator	description	category	IFC	ubiquity
v2clprptym	Property rights for men	Property Rights	-0.263	106
v2csptrcpt	Civil society participatory environment	Civil Society	-0.266	108
v2lgfunds	Legislature controls resources	Legislative Constraints	-0.266	96
v2jupack	Court packing	Judicial Independence	-0.272	102
v2jupurge	Purges of the judiciary	Judicial Independence	-0.305	95
v2clstown	State ownership of economy	Civil Liberties	-0.376	106
v2pepwrgen	Power distributed by gender	Equality	-0.392	92
v2elpeace	Election other voting irregularities	Electoral Integrity	-0.393	88
v2dlcommon	Common good justification	Deliberation	-0.428	102
v2lgdsadlo	Legislature disbands/self-limits	Legislative Constraints	-0.505	91
v2pepwrses	Power distributed by socioeconomic position	Equality	-0.583	100
v2jureview	Judicial review	Judicial Independence	-0.740	105
v2strenadm	Remuneration in state administration	State Capacity	-1.266	104
v2x_genpp	Women's civil society participation index	Equality	-1.268	125
v2svsterr	State authority over territory	State Capacity	-1.421	112
v2x_elecoff	Elected officials index	Electoral Integrity	-1.504	145

indicator	description	category	IFC	ubiquity
v2lgello	Legislature elections are free/fair	Legislative Constraints	-1.728	142
v2csantimv	CSO anti-system movements	Civil Society	-2.861	95
v2x_pubcorr	Public sector corruption index	State Capacity	-5.321	96
v2x_corr	Political corruption index	Anti-Corruption	-5.492	92

```
ggplot(pci_df, aes(x = reorder(category, IFC, median), y = IFC, fill = category)) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  coord_flip() +
  labs(
    title = "Institutional Feature Complexity by Category",
    x = NULL, y = "Institutional Feature Complexity (IFC)"
  ) +
  theme_minimal(base_size = 13)
```



Proximity and Relatedness

Institution proximity

We calculate the proximity between institutional features using the conditional probability approach from Hidalgo et al. (2007):

$$\phi_{pp'} = \min (P(p|p'), P(p'|p))$$

where $P(p|p')$ is the probability that a country has feature p given that it has feature p' . This measures how likely two features are to co-occur.

```
n_features <- ncol(Mcp)
phi_pp <- matrix(0, n_features, n_features)
rownames(phi_pp) <- colnames(Mcp)
colnames(phi_pp) <- colnames(Mcp)

for (i in 1:n_features) {
  for (j in i:n_features) {
    if (i == j) {
      phi_pp[i, j] <- 1
    } else {
      co_occur <- sum(Mcp[, i] & Mcp[, j])
      p_given_j <- if (kp0[j] > 0) co_occur / kp0[j] else 0
      p_given_i <- if (kp0[i] > 0) co_occur / kp0[i] else 0
      phi_pp[i, j] <- min(p_given_i, p_given_j)
      phi_pp[j, i] <- phi_pp[i, j]
    }
  }
}

cat("Institution proximity matrix:", nrow(phi_pp), "x", ncol(phi_pp), "\n")

## Institution proximity matrix: 96 x 96
cat("Mean proximity:", round(mean(phi_pp[upper.tri(phi_pp)]), 3), "\n")

## Mean proximity: 0.706
```

Country proximity

Similarly, we compute proximity between countries:

$$\phi_{cc'} = \min(P(c|c'), P(c'|c))$$

```
n_countries <- nrow(Mcp)
phi_cc <- matrix(0, n_countries, n_countries)
rownames(phi_cc) <- rownames(Mcp)
colnames(phi_cc) <- rownames(Mcp)

for (i in 1:n_countries) {
  for (j in i:n_countries) {
    if (i == j) {
      phi_cc[i, j] <- 1
    } else {
      co_occur <- sum(Mcp[i, ] & Mcp[j, ])
      p_given_j <- if (kc0[j] > 0) co_occur / kc0[j] else 0
      p_given_i <- if (kc0[i] > 0) co_occur / kc0[i] else 0
      phi_cc[i, j] <- min(p_given_i, p_given_j)
      phi_cc[j, i] <- phi_cc[i, j]
    }
  }
}
```

Institution Space (Network)

The institution space is a network where nodes are institutional features and edges connect features that tend to co-occur across countries.

```
# Threshold for edges: use a percentile-based cutoff
threshold_p <- quantile(phi_pp[upper.tri(phi_pp)], 0.75)

# Build adjacency matrix
adj_pp <- phi_pp
adj_pp[adj_pp < threshold_p] <- 0
diag(adj_pp) <- 0

# Create igraph object
g_inst <- graph_from_adjacency_matrix(adj_pp, mode = "undirected", weighted = TRUE)

# Add attributes
V(g_inst)$category <- indicator_info$category[match(V(g_inst)$name, indicator_info$code)]
V(g_inst)$complexity <- pci_eigen[V(g_inst)$name]
V(g_inst)$ubiquity <- kp0[V(g_inst)$name]

cat("Institution space: ", vcount(g_inst), "nodes,", ecoun(g_inst), "edges\n")

## Institution space: 96 nodes, 1145 edges
cat("Mean degree:", round(mean(degree(g_inst)), 1), "\n")
```

Mean degree: 23.9

Country Space (Network)

```
threshold_c <- quantile(phi_cc[upper.tri(phi_cc)], 0.80)

adj_cc <- phi_cc
adj_cc[adj_cc < threshold_c] <- 0
diag(adj_cc) <- 0

g_country <- graph_from_adjacency_matrix(adj_cc, mode = "undirected", weighted = TRUE)

# Add region info
region_lookup <- vdem_filtered %>%
  select(country_name, e_regionpol_6C) %>%
  deframe()

region_labels <- c(
  "1" = "Eastern Europe & Central Asia",
  "2" = "Latin America & Caribbean",
  "3" = "Middle East & North Africa",
  "4" = "Sub-Saharan Africa",
  "5" = "Western Europe & North America",
  "6" = "East Asia & Pacific"
)

V(g_country)$region <- region_labels[as.character(
  region_lookup[V(g_country)$name]
```

```

)]
V(g_country)$ICI <- eci_eigen[V(g_country)$name]

cat("Country space:", vcount(g_country), "nodes,", ecount(g_country), "edges\n")

## Country space: 179 nodes, 3191 edges

```

UMAP Visualizations

Institution Space (UMAP)

We use UMAP to embed the institution proximity matrix into 2D for visualization. UMAP preserves local structure well, making it ideal for revealing clusters of related institutions.

```

# Convert proximity to distance, use cmdscale (MDS) as fallback since uwot
# doesn't support precomputed distance matrices well on all versions
inst_dist <- as.dist(1 - phi_pp)

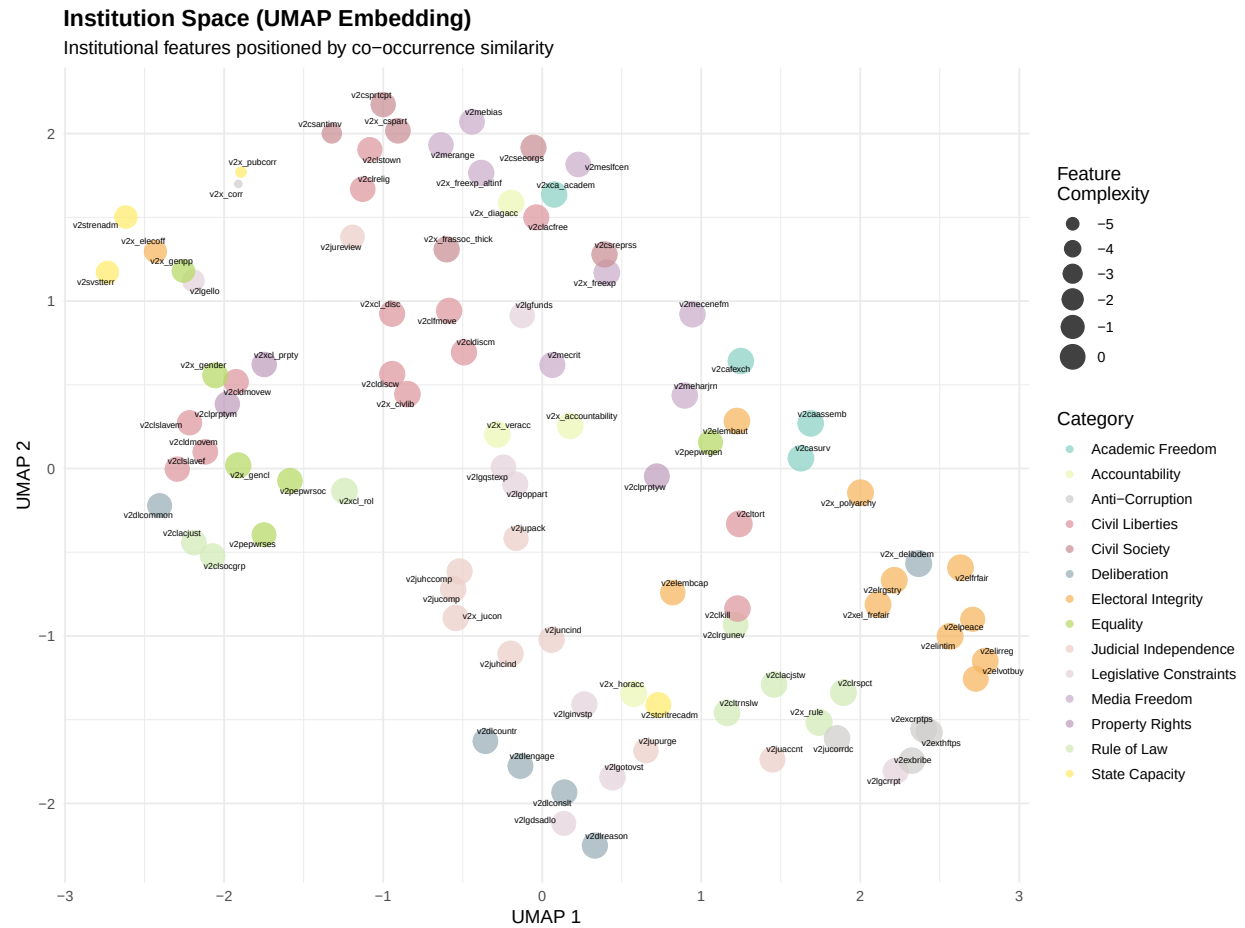
# Use MDS first, then UMAP on the MDS coordinates for robust embedding
set.seed(42)
mds_inst <- cmdscale(inst_dist, k = min(10, ncol(Mcp) - 1))
umap_inst <- umap(mds_inst, n_neighbors = 15, min_dist = 0.1, n_epochs = 500)

umap_inst_df <- tibble(
  indicator = colnames(Mcp),
  UMAP1 = umap_inst[, 1],
  UMAP2 = umap_inst[, 2],
  ubiquity = kp0,
  IFC = pci_eigen
) %>%
  left_join(indicator_info, by = c("indicator" = "code"))

# Color palette for categories
n_cats <- length(unique(umap_inst_df$category))
cat_colors <- colorRampPalette(brewer.pal(12, "Set3"))(n_cats)

ggplot(umap_inst_df, aes(x = UMAP1, y = UMAP2)) +
  geom_point(aes(color = category, size = IFC), alpha = 0.75) +
  geom_text_repel(
    aes(label = indicator),
    size = 2.2, max.overlaps = 20, segment.alpha = 0.3
  ) +
  scale_size_continuous(range = c(2, 8), name = "Feature\nComplexity") +
  scale_color_manual(values = cat_colors, name = "Category") +
  labs(
    title = "Institution Space (UMAP Embedding)",
    subtitle = "Institutional features positioned by co-occurrence similarity",
    x = "UMAP 1", y = "UMAP 2"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold", size = 16)
  )

```



Country Space (UMAP)

```
set.seed(42)
country_dist <- as.dist(1 - phi_cc)
mds_country <- cmdscale(country_dist, k = min(10, nrow(Mcp) - 1))
umap_country <- umap(mds_country, n_neighbors = 15, min_dist = 0.2, n_epochs = 500)

umap_country_df <- tibble(
  country = rownames(Mcp),
  UMAP1 = umap_country[, 1],
  UMAP2 = umap_country[, 2],
  ICI = eci_eigen,
  diversity = kc0
)

# Add region
umap_country_df <- umap_country_df %>%
  mutate(
    region = region_labels[as.character(region_lookup[country])],
    region = replace_na(region, "Other")
  )
```

```

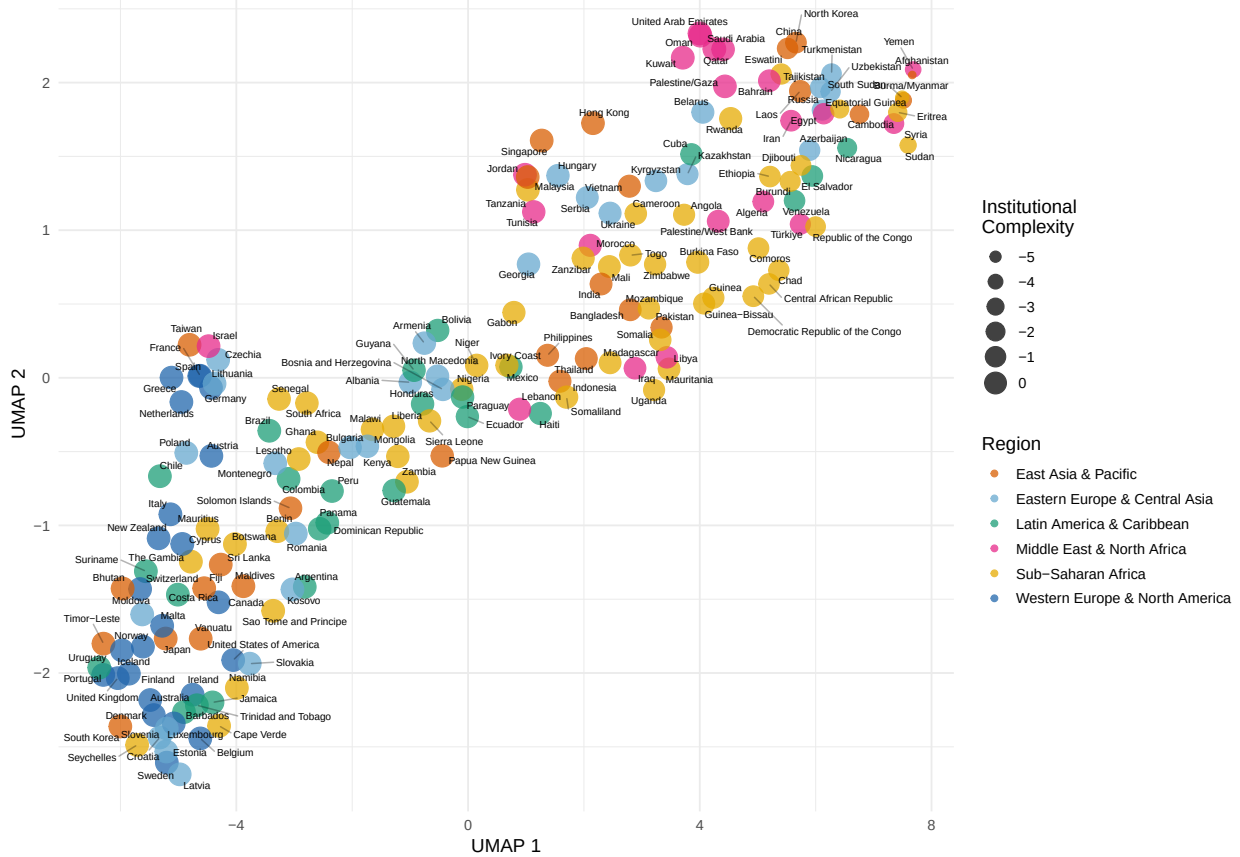
region_colors <- c(
  "Western Europe & North America" = "#2166AC",
  "Eastern Europe & Central Asia" = "#67A9CF",
  "Latin America & Caribbean" = "#1B9E77",
  "East Asia & Pacific" = "#D95F02",
  "Middle East & North Africa" = "#E7298A",
  "Sub-Saharan Africa" = "#E6AB02",
  "Other" = "grey50"
)

ggplot(umap_country_df, aes(x = UMAP1, y = UMAP2)) +
  geom_point(aes(color = region, size = ICI), alpha = 0.75) +
  geom_text_repel(
    aes(label = country),
    size = 2.5, max.overlaps = 25, segment.alpha = 0.3
  ) +
  scale_size_continuous(range = c(2, 7), name = "Institutional\nComplexity") +
  scale_color_manual(values = region_colors, name = "Region") +
  labs(
    title = "Country Space (UMAP Embedding)",
    subtitle = "Countries positioned by institutional portfolio similarity",
    x = "UMAP 1", y = "UMAP 2"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold", size = 16)
  )

```


Country Space (UMAP Embedding)

Countries positioned by institutional portfolio similarity



Additional Visualizations

Heatmap of the Mcp Matrix

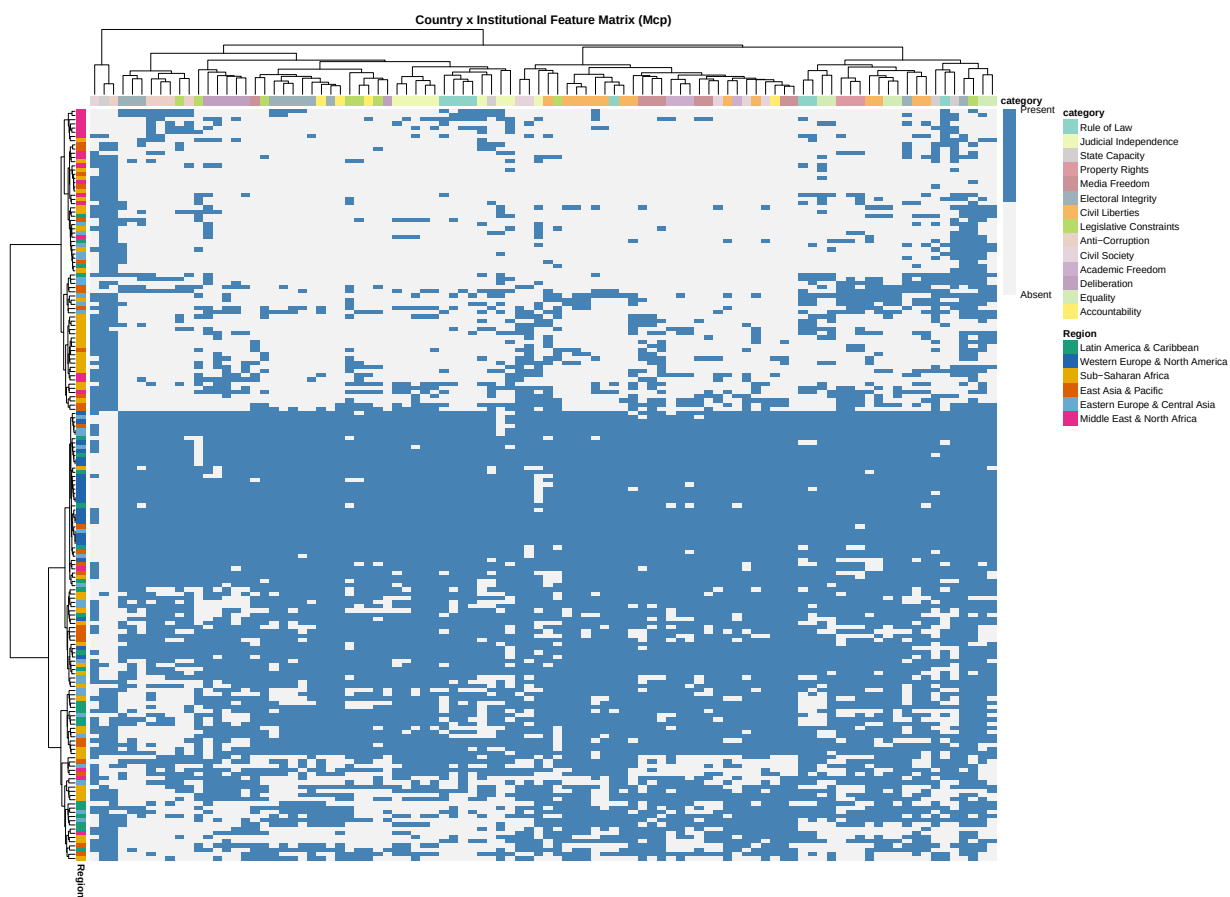
```
# Cluster and plot heatmap
# Add category annotations for columns
col_annotation <- indicator_info %>%
  filter(code %in% colnames(Mcp)) %>%
  select(code, category) %>%
  column_to_rownames("code")

# Add region annotations for rows
row_annotation <- tibble(
  country = rownames(Mcp),
  Region = region_labels[as.character(region_lookup[rownames(Mcp)])]
) %>%
  mutate(Region = replace_na(Region, "Other")) %>%
  column_to_rownames("country")

# Annotation colors
ann_colors <- list(
  category = setNames(cat_colors, unique(col_annotation$category)),
  Region = region_colors[unique(row_annotation$Region)]
)
```

```
)

pheatmap(
  Mcp,
  color = c("grey95", "steelblue"),
  legend_breaks = c(0, 1),
  legend_labels = c("Absent", "Present"),
  annotation_col = col_annotation,
  annotation_row = row_annotation,
  annotation_colors = ann_colors,
  show_rownames = FALSE,
  show_colnames = FALSE,
  clustering_method = "ward.D2",
  main = "Country x Institutional Feature Matrix (Mcp)",
  fontsize = 8
)
```



ICI vs GDP per Capita

```
# Get GDP per capita from World Development Indicators
gdp_data <- WDI(
  indicator = "NY.GDP.PCAP.PP.KD",
  start = latest_year - 1,
```

```

    end = latest_year,
    extra = TRUE
  ) %>%
  filter(!is.na(NY.GDP.PCAP.PP.KD)) %>%
  group_by(country) %>%
  slice_max(year, n = 1) %>%
  ungroup() %>%
  select(country, iso3c, gdp_pc = NY.GDP.PCAP.PP.KD)

# Match to our country names using countrycode
eci_gdp <- eci_df %>%
  mutate(
    iso3c = countrycode(country, "country.name", "iso3c",
                        warn = FALSE)
  ) %>%
  left_join(gdp_data, by = "iso3c") %>%
  filter(!is.na(gdp_pc))

cat("Countries matched with GDP data:", nrow(eci_gdp), "\n")

## Countries matched with GDP data: 165

# Add regions
eci_gdp <- eci_gdp %>%
  mutate(
    region = region_labels[as.character(
      region_lookup[country.x]
    )],
    region = replace_na(region, "Other")
  )

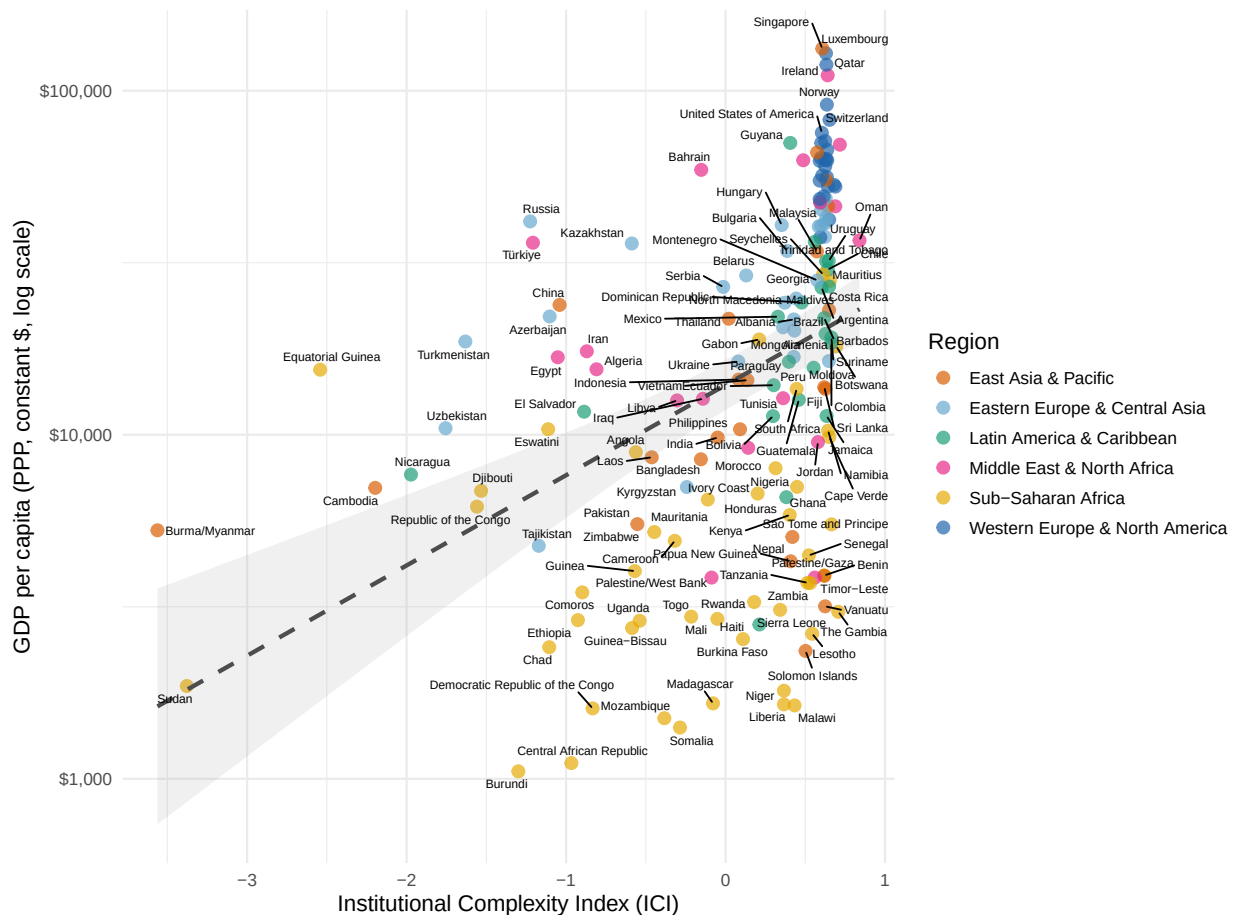
cor_gdp <- cor(eci_gdp$ICI, log(eci_gdp$gdp_pc), use = "complete.obs")

ggplot(eci_gdp, aes(x = ICI, y = gdp_pc)) +
  geom_point(aes(color = region), size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", color = "grey30", linetype = "dashed", se = TRUE, alpha = 0.15) +
  geom_text_repel(
    aes(label = country.x),
    size = 2.5, max.overlaps = 15
  ) +
  scale_y_log10(labels = scales::dollar_format()) +
  scale_color_manual(values = region_colors, name = "Region") +
  labs(
    title = "Institutional Complexity vs GDP per Capita",
    subtitle = sprintf("Correlation with log GDP: %.2f", cor_gdp),
    x = "Institutional Complexity Index (ICI)",
    y = "GDP per capita (PPP, constant $, log scale)"
  ) +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold", size = 16))

```

Institutional Complexity vs GDP per Capita

Correlation with log GDP: 0.41



ICI vs Liberal Democracy Index

```
# Get liberal democracy index
libdem <- vdem_raw %>%
  filter(year == latest_year) %>%
  select(country_name, v2x_libdem) %>%
  filter(!is.na(v2x_libdem))

eci_libdem <- eci_df %>%
  left_join(libdem, by = c("country" = "country_name")) %>%
  filter(!is.na(v2x_libdem)) %>%
  mutate(
    region = region_labels[as.character(region_lookup[country])],
    region = replace_na(region, "Other")
  )

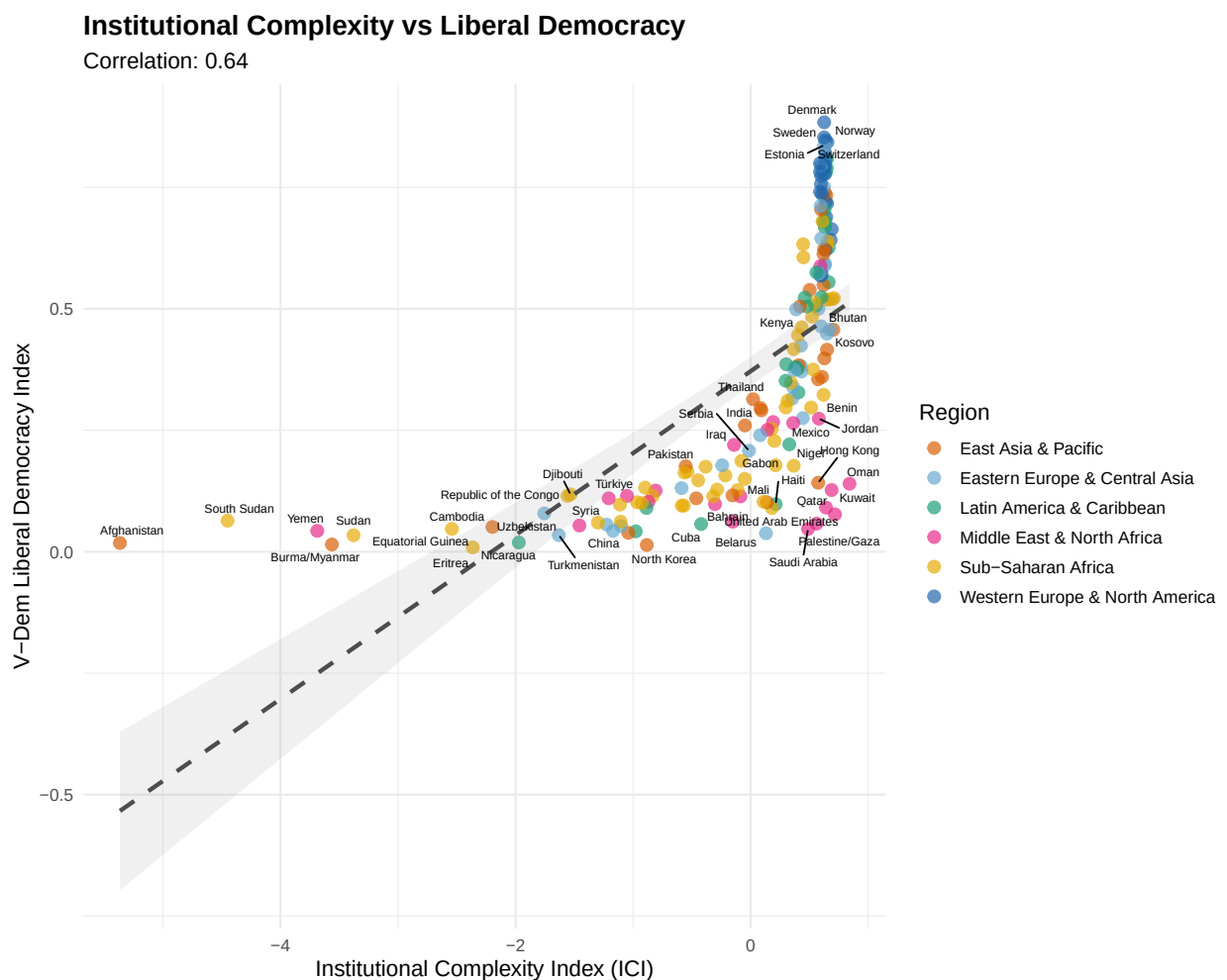
cor_libdem <- cor(eci_libdem$ICI, eci_libdem$v2x_libdem, use = "complete.obs")

ggplot(eci_libdem, aes(x = ICI, y = v2x_libdem)) +
  geom_point(aes(color = region), size = 3, alpha = 0.7) +
  geom_smooth(method = "lm", color = "grey30", linetype = "dashed", se = TRUE, alpha = 0.15) +
```

```

geom_text_repel(
  aes(label = country),
  size = 2.5, max.overlaps = 15
) +
scale_color_manual(values = region_colors, name = "Region") +
labs(
  title = "Institutional Complexity vs Liberal Democracy",
  subtitle = sprintf("Correlation: %.2f", cor_libdem),
  x = "Institutional Complexity Index (ICI)",
  y = "V-Dem Liberal Democracy Index"
) +
theme_minimal(base_size = 13) +
theme(plot.title = element_text(face = "bold", size = 16))

```



ICI by World Region

```

eci_regions <- eci_df %>%
  mutate(
    region = region_labels[as.character(region_lookup[country])],
    region = replace_na(region, "Other")
  )

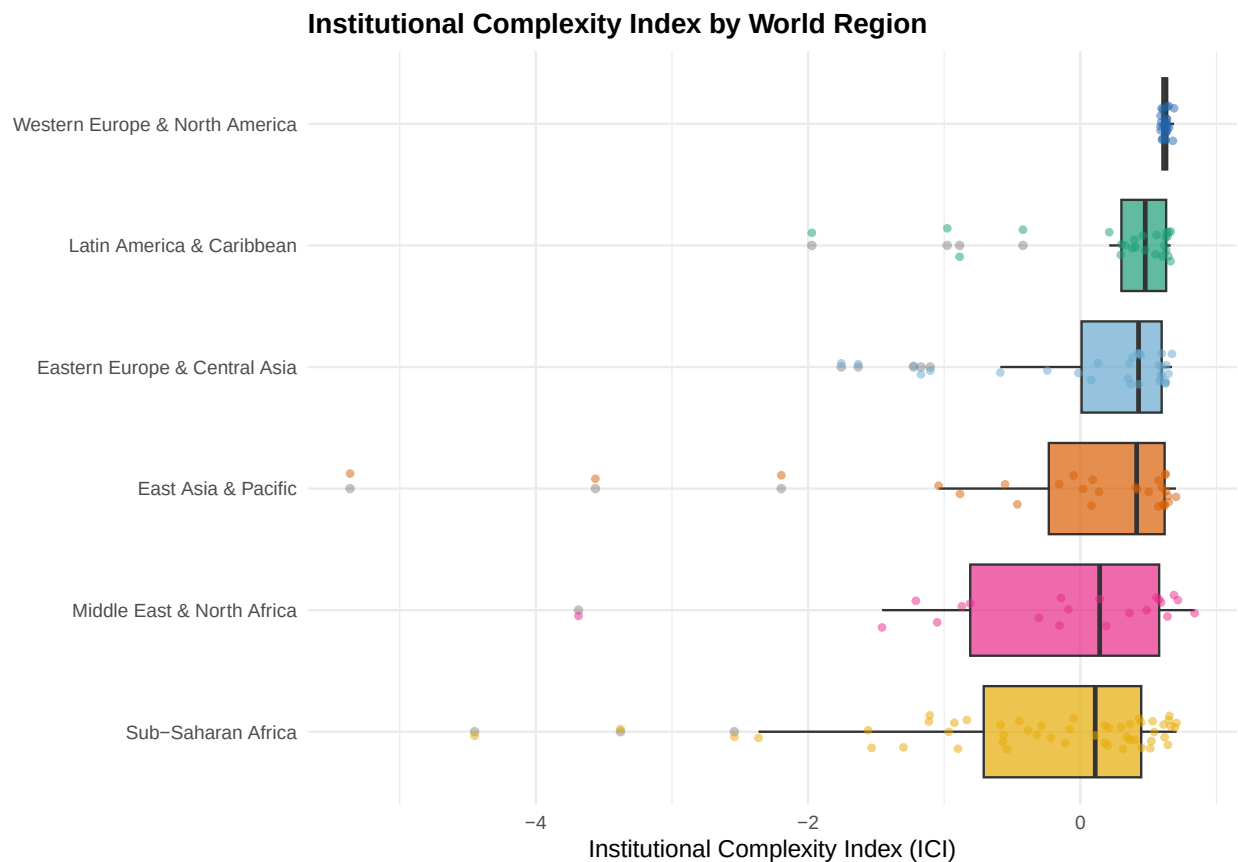
```

```

) %>%
  filter(region != "Other")

ggplot(eci_regions, aes(x = reorder(region, ICI, median), y = ICI, fill = region)) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE, outlier.alpha = 0.3) +
  geom_jitter(aes(color = region), width = 0.15, alpha = 0.5, size = 1.5, show.legend = FALSE) +
  coord_flip() +
  scale_fill_manual(values = region_colors) +
  scale_color_manual(values = region_colors) +
  labs(
    title = "Institutional Complexity Index by World Region",
    x = NULL, y = "Institutional Complexity Index (ICI)"
  ) +
  theme_minimal(base_size = 14) +
  theme(plot.title = element_text(face = "bold", size = 16))

```



Discussion

What does institutional complexity tell us?

The analogy between economic and institutional complexity reveals several patterns:

High-complexity countries tend to be established democracies with deep, interlocking institutional frameworks. These countries don't just have individual strong institutions – they have *rare combinations* of

institutions that reinforce each other. Just as economically complex countries produce products that require many complementary capabilities, institutionally complex countries maintain governance features that require deep state capacity, civic culture, and historical development.

The institution space reveals how governance features cluster. Features that are close together in the UMAP embedding tend to co-occur – they represent institutional “packages” that countries adopt together. The structure of this space suggests that some institutional reforms may be easier to adopt once neighboring capabilities are in place.

Feature complexity is particularly informative. The most “complex” institutional features – those found primarily in institutionally sophisticated countries – tend to involve subtle aspects of governance: rigorous and impartial public administration, judicial accountability, freedom from self-censorship in media, and deliberative quality of political discourse. These are features that require not just formal rules but deeply embedded norms and practices.

Caveats

Several important limitations:

1. **V-Dem data reflects expert assessments**, not objective measures. The binarization choices affect results significantly.
2. **The analogy has limits.** Countries don’t “choose” institutions the way firms choose products to export. Institutional development is path-dependent, often violent, and constrained by geography, history, and international power relations.
3. **Correlation with GDP and democracy is expected** given that many V-Dem indicators directly measure democratic quality. The interesting question is whether the *complexity* measure adds information beyond simple additive indices.
4. **The method is sensitive to indicator selection.** A different set of indicators would produce different complexity rankings. Our selection, while broad, reflects choices about what counts as an “institution.”
5. **Cross-sectional analysis only.** A natural extension would track institutional complexity over time to study how countries build (or lose) institutional sophistication.

Future directions

- Panel analysis: how does institutional complexity evolve over time?
- Predictive power: does ICI predict future economic growth or democratic stability beyond what simpler indices capture?
- Institutional “product space” analysis: which institutional upgrades are most feasible for countries at different positions in the institution space?
- Comparison with actual Economic Complexity Index: are economically and institutionally complex countries the same?

Analysis generated on 2026-04-20. Data from V-Dem v16.0 and World Development Indicators.